Lecture Notes

Lecture 8

Orthonormal Bases, Gram–Schmidt, and Riesz

Lecture 7 ended with a question: can we always find a basis of mutually orthogonal unit vectors, and expand any state in it? This lecture answers yes. The *Gram–Schmidt procedure* manufactures an *orthonormal basis* from any basis, and in such a basis everything becomes simple: coordinates are just inner products, $v = \sum_k \langle e_k, v\rangle\, e_k$; the norm is the Pythagorean sum $\|v\|^2 = \sum_k |\langle e_k, v\rangle|^2$; and the identity resolves as $\sum_k |e_k\rangle\langle e_k| = I$. We then use orthonormality to solve the *best-approximation* problem (orthogonal projection finds the closest point of a subspace) and to prove the *Riesz representation theorem*, which says every bra is the partner of a unique ket. These are the daily tools of quantum mechanics: expanding a state in an energy eigenbasis, computing amplitudes, and projecting onto measurement outcomes.

Learning objectives

By the end of this lecture you will be able to:

- Expand vectors in an orthonormal basis, with coordinates as inner products.
- Carry out the Gram–Schmidt procedure to build an orthonormal basis.
- Compute orthogonal projections and solve best-approximation / least-squares problems.
- State the Riesz representation theorem and link it to the bra–ket duality.

1 Orthonormal bases

Definition 8.1 (Orthonormal list and basis). A list $e_1, \dots, e_m$ is *orthonormal* if $\langle e_j, e_k\rangle = \delta_{jk}$ (each is a unit vector, distinct ones orthogonal). An orthonormal list that is also a basis is an *orthonormal basis* (ONB).

Lemma 8.2 (Orthonormal lists are independent). Every orthonormal list is linearly independent.

Proof. If $\sum_k c_k e_k = 0$, pairing with e_j gives $0 = \langle e_j, \sum_k c_k e_k\rangle = \sum_k c_k \delta_{jk} = c_j$. So every coefficient vanishes. ∎

The point of an ONB is that coordinates require no linear system—they are inner products.

Theorem 8.3 (Expansion in an orthonormal basis). If $e_1, \dots, e_n$ is an orthonormal basis of V, then for every $v \in V$

$$v = \sum_{k=1}^{n} \langle e_k, v\rangle\, e_k, \qquad \|v\|^2 = \sum_{k=1}^{n} |\langle e_k, v\rangle|^2.$$

The second identity is *Parseval's relation.*

Proof. Write $v = \sum_k c_k e_k$ (possible since the e_k are a basis). Pairing with e_j and using orthonormality, $\langle e_j, v\rangle = \sum_k c_k \delta_{jk} = c_j$, which gives the first formula. Then $\|v\|^2 = \langle v, v\rangle = \sum_{j,k} \overline{c_j} c_k \langle e_j, e_k\rangle = \sum_k |c_k|^2$, the second. ∎

Corollary 8.4 (Inner products from coordinates). In an orthonormal basis, $\langle u, v\rangle = \sum_k \overline{\langle e_k, u\rangle}\,\langle e_k, v\rangle$. Once vectors are expanded in an ONB, every inner product is the standard sum $\sum_k \overline{a_k} b_k$ of their coordinates—which is why $\mathbb{C}^n$ with its standard product is the universal model: any n-dimensional complex inner product space is isometric to it.

Proof. Expand $u = \sum_j \langle e_j, u\rangle e_j$ and $v = \sum_k \langle e_k, v\rangle e_k$; then

$$\langle u, v\rangle = \sum_{j,k} \overline{\langle e_j, u\rangle}\langle e_k, v\rangle\langle e_j, e_k\rangle = \sum_k \overline{\langle e_k, u\rangle}\langle e_k, v\rangle. \quad ∎$$

Remark 8.5 (Completeness relation). In Dirac notation the expansion reads

$$|v\rangle = \sum_k |e_k\rangle\langle e_k, v\rangle = \Big(\sum_k |e_k\rangle\langle e_k|\Big)|v\rangle,$$

so an orthonormal basis satisfies the *completeness relation* $\sum_k |e_k\rangle\langle e_k| = I$. Inserting this resolution of the identity between symbols is the basic move of quantum-mechanical calculation.

Example 8.6 (Three orthonormal bases). The standard basis of $\mathbb{C}^n$ is orthonormal. So is the qubit pair $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ of Lecture 7. On the space of finite Fourier series, the functions $\frac{1}{\sqrt{2\pi}}, \frac{1}{\sqrt{\pi}}\cos kx, \frac{1}{\sqrt{\pi}}\sin kx$ are orthonormal for the inner product $\langle f, g\rangle = \int_{-\pi}^{\pi} \overline{f} g\, dx$—the basis behind Fourier analysis. ◁

Example 8.7 (Coordinates as inner products). Expand $v = (3, 4) \in \mathbb{R}^2$ in the ONB $e_1 = \frac{1}{\sqrt{2}}(1, 1)$, $e_2 = \frac{1}{\sqrt{2}}(1, -1)$. The coordinates are $\langle e_1, v\rangle = \frac{7}{\sqrt{2}}$ and $\langle e_2, v\rangle = -\frac{1}{\sqrt{2}}$, so $v = \frac{7}{\sqrt{2}}e_1 - \frac{1}{\sqrt{2}}e_2$, and Parseval checks: $\frac{49}{2} + \frac{1}{2} = 25 = \|v\|^2$. No linear system was solved. ◁

Example 8.8 (A Fourier coefficient). In the Fourier basis above, the coordinate of a signal f along $\frac{1}{\sqrt{\pi}}\sin kx$ is $\frac{1}{\sqrt{\pi}}\int_{-\pi}^{\pi} \sin(kx) f(x)\, dx$, the kth Fourier sine coefficient. Parseval then equates the "energy" $\|f\|^2$ with the sum of squared coefficients—the foundation of spectral analysis. ◁

Example 8.9 (A complex orthonormal basis). On $\mathbb{C}^2$ the vectors $e_1 = \frac{1}{\sqrt{5}}(2, i)$ and $e_2 = \frac{1}{\sqrt{5}}(1, -2i)$ are orthonormal: $\|e_1\|^2 = \frac{1}{5}(4 + 1) = 1$, $\|e_2\|^2 = \frac{1}{5}(1 + 4) = 1$, and $\langle e_1, e_2\rangle = \frac{1}{5}\big(\overline{2}\cdot 1 + \overline{i}\cdot(-2i)\big) = \frac{1}{5}(2 - 2) = 0$. The conjugation in the first slot is exactly what produces the cancellation. ◁

2 The Gram–Schmidt procedure

Theorem 8.10 (Gram–Schmidt). Every finite-dimensional inner product space has an orthonormal basis. Concretely, from any basis $v_1, \dots, v_n$ one builds an orthonormal basis $e_1, \dots, e_n$ with $\mathrm{span}(e_1, \dots, e_k) = \mathrm{span}(v_1, \dots, v_k)$ for every k, by

$$e_1 = \frac{v_1}{\|v_1\|}, \qquad e_k = \frac{w_k}{\|w_k\|}, \qquad w_k = v_k - \sum_{j<k} \langle e_j, v_k \rangle \, e_j.$$

Proof. We induct on k, assuming $e_1, \dots, e_{k-1}$ orthonormal with the right span. The numerator $w_k = v_k - \sum_{j<k} \langle e_j, v_k \rangle e_j$ is nonzero, else v_k would lie in $\mathrm{span}(v_1, \dots, v_{k-1})$, contradicting independence. For $i < k$,

$$\langle e_i, w_k \rangle = \langle e_i, v_k \rangle - \sum_{j<k} \langle e_j, v_k \rangle \langle e_i, e_j \rangle = \langle e_i, v_k \rangle - \langle e_i, v_k \rangle = 0,$$

so $e_k = w_k / \|w_k\|$ is a unit vector orthogonal to the earlier ones. Since v_k and e_k differ only by a scalar and a combination of $e_1, \dots, e_{k-1}$, the spans agree. After n steps we have n orthonormal vectors, a basis by Lemma 8.2. ■

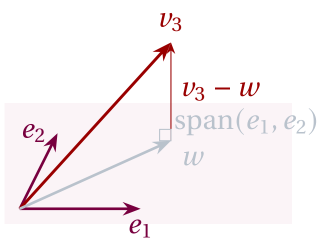

Figure 1: The third Gram–Schmidt step: subtract from v_3 its projection $w = \langle e_1, v_3 \rangle e_1 + \langle e_2, v_3 \rangle e_2$ onto $\mathrm{span}(e_1, e_2)$; the remainder $v_3 - w$ is orthogonal to the plane and is normalized to give e_3.

Example 8.11 (Gram–Schmidt in $\mathbb{R}^3$). Apply the procedure to $v_1 = (1, 1, 0)$, $v_2 = (2, 0, 1)$, $v_3 = (1, -2, -2)$. First $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$. Then

$$v_2 - \langle e_1, v_2 \rangle e_1 = (2, 0, 1) - (1, 1, 0) = (1, -1, 1), \qquad e_2 = \tfrac{1}{\sqrt{3}}(1, -1, 1).$$

Finally $v_3 - \langle e_1, v_3 \rangle e_1 - \langle e_2, v_3 \rangle e_2 = \frac{7}{6}(1, -1, -2)$, so $e_3 = \frac{1}{\sqrt{6}}(1, -1, -2)$. One checks $\langle e_i, e_j \rangle = \delta_{ij}$. ◁

Example 8.12 (Verifying orthonormality). For the output of Example 8.11, the off-diagonal checks are $\langle e_1, e_2 \rangle = \frac{1}{\sqrt{6}}(1 - 1 + 0) = 0$, $\langle e_1, e_3 \rangle = \frac{1}{\sqrt{12}}(1 - 1 + 0) = 0$, and $\langle e_2, e_3 \rangle = \frac{1}{\sqrt{18}}(1 + 1 - 2) = 0$, while each $\|e_i\| = 1$. Confirming $\langle e_i, e_j \rangle = \delta_{ij}$ directly is always the way to validate a Gram–Schmidt computation. ◁

Example 8.13 (Gram–Schmidt over $\mathbb{C}$). Orthonormalize $v_1 = (1, i)$, $v_2 = (1, 0)$ in $\mathbb{C}^2$. First $e_1 = \frac{1}{\sqrt{2}}(1, i)$. Then $\langle e_1, v_2 \rangle = \frac{1}{\sqrt{2}}\overline{1} \cdot 1 = \frac{1}{\sqrt{2}}$, so $v_2 - \langle e_1, v_2 \rangle e_1 = (1, 0) - \frac{1}{2}(1, i) = \frac{1}{2}(1, -i)$, which normalizes to $e_2 = \frac{1}{\sqrt{2}}(1, -i)$. A check gives $\langle e_1, e_2 \rangle = \frac{1}{2}(1 + i^2) = 0$; the conjugations must be tracked carefully over $\mathbb{C}$. ◁

Example 8.14 (Legendre polynomials). On $\mathcal{P}_2$ with $\langle p, q \rangle = \int_{-1}^{1} \overline{p}\, q \, dx$, orthonormalize $1, x, x^2$. Since $\int_{-1}^{1} 1 \, dx = 2$, $p_0 = \frac{1}{\sqrt{2}}$. As $\langle p_0, x \rangle = 0$, x is already orthogonal to p_0; normalizing, $p_1 = \sqrt{\frac{3}{2}}\, x$. For the quadratic, $x^2 - \langle p_0, x^2 \rangle p_0 - \langle p_1, x^2 \rangle p_1 = x^2 - \frac{1}{3}$, and normalizing gives

$$p_2 = \sqrt{\tfrac{5}{8}}\, (3x^2 - 1).$$

These are the first *Legendre polynomials*, ubiquitous in electrostatics and the quantum theory of angular momentum. ◁

Remark 8.15 (Gram determinant and volume). The determinant of the Gram matrix $G_{jk} = \langle v_j, v_k\rangle$ equals the squared volume of the parallelepiped spanned by $v_1, \ldots, v_m$, and it is nonzero exactly when the vectors are independent (Lecture 7). Gram–Schmidt computes the same content step by step: the norms $\|w_k\|$ are the successive "heights" whose product is that volume.

Example 8.16 (Orthonormal basis of a plane). The plane $U = \{(x, y, z) : x - y - 2z = 0\} \subset \mathbb{R}^3$ contains the independent vectors $v_1 = (2, 0, 1)$ and $v_2 = (3, 1, 1)$. Gram–Schmidt gives $e_1 = \frac{1}{\sqrt{5}}(2, 0, 1)$, and from $v_2 - \langle e_1, v_2\rangle e_1 = (3, 1, 1) - \frac{7}{5}(2, 0, 1) = \frac{1}{5}(1, 5, -2)$, $e_2 = \frac{1}{\sqrt{30}}(1, 5, -2)$. These orthonormal vectors span U. ◁

> **Corollary 8.17 (Extending to an orthonormal basis).** Every orthonormal list in a finite-dimensional V extends to an orthonormal basis of V.

Proof. Extend the orthonormal list to a basis (Lecture 2), then run Gram–Schmidt; the already-orthonormal vectors are unchanged, and the rest become an orthonormal completion. ∎

Remark 8.18 (The QR factorization). Collecting the relations $v_k = \sum_{j\le k} r_{jk} e_j$ into matrices says $A = QR$: any matrix with independent columns factors as a matrix Q with orthonormal columns times an upper-triangular R. This is the numerically standard route to least-squares problems and to eigenvalue algorithms.

Remark 8.19 (Orthonormal bases are far from unique). Gram–Schmidt depends on the ordering of its inputs and on a choice of sign (or phase) at each normalization, so a space carries infinitely many orthonormal bases. Any two are related by a *unitary* change of basis—the inner-product-preserving operators of Lecture 9.

Remark 8.20 (Numerical caution). In floating-point arithmetic the classical formula loses orthogonality through accumulated rounding; the *modified* Gram–Schmidt, which subtracts each projection as soon as it is computed rather than all at once, is the stable variant used in practice. The mathematics is identical, the numerics are not.

3 Orthogonal projection and best approximation

Let U be a subspace of V with orthonormal basis $e_1, \ldots, e_m$. The *orthogonal projection* onto U is

$$P_U v = \sum_{k=1}^{m} \langle e_k, v\rangle\, e_k.$$

> **Theorem 8.21 (Orthogonal decomposition).** For each $v \in V$, $v - P_U v \in U^\perp$, and $V = U \oplus U^\perp$. Thus P_U is the projection onto U along $U^\perp$.

Proof. For each i, $\langle e_i, v - P_U v\rangle = \langle e_i, v\rangle - \sum_k \langle e_k, v\rangle\langle e_i, e_k\rangle = \langle e_i, v\rangle - \langle e_i, v\rangle = 0$, so $v - P_U v$ is orthogonal to every e_i and hence to U. Then $v = P_U v + (v - P_U v)$ exhibits v as a sum of a vector in U and one in $U^\perp$; since $U \cap U^\perp = \{0\}$ (a vector orthogonal to itself is 0), the sum is direct. ∎

Proposition 8.22 (Dimension of the orthogonal complement). For a subspace U of a finite-dimensional V, $\dim U + \dim U^\perp = \dim V$, and $(U^\perp)^\perp = U$.

Proof. Take an orthonormal basis $e_1, \dots, e_m$ of U and extend it to an orthonormal basis $e_1, \dots, e_n$ of V (Corollary 8.17). The extra vectors $e_{m+1}, \dots, e_n$ lie in $U^\perp$ and span it—any $v \in U^\perp$ has zero coordinates along $e_1, \dots, e_m$—so $\dim U^\perp = n - m$. For the second claim, $U \subseteq (U^\perp)^\perp$ straight from the definition: every $u \in U$ is orthogonal to every vector of $U^\perp$. Applying the dimension identity to $U^\perp$ gives $\dim(U^\perp)^\perp = n - (n - m) = m = \dim U$, and a subspace contained in another of the same finite dimension equals it, so $(U^\perp)^\perp = U$. ∎

Example 8.23 (Computing an orthogonal complement). For $U = \operatorname{span}\big((1, 2, 3, -4)\big) \subset \mathbb{R}^4$, the complement $U^\perp$ is the solution set of $\langle (1, 2, 3, -4), v\rangle = v_1 + 2v_2 + 3v_3 - 4v_4 = 0$, a three-dimensional hyperplane—consistent with $\dim U + \dim U^\perp = 1 + 3 = 4$. Reading off the free-variable solutions gives a basis $(-2, 1, 0, 0)$, $(-3, 0, 1, 0)$, $(4, 0, 0, 1)$ of $U^\perp$. ◁

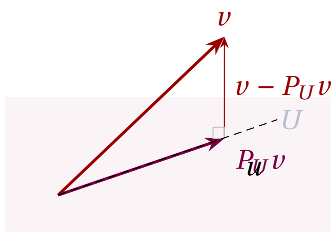

Figure 2: Best approximation: of all points $u \in U$, the orthogonal projection $P_U v$ is closest to v, because the error $v - P_U v$ is perpendicular to U.

Theorem 8.24 (Best approximation). For every $u \in U$, $\|v - P_U v\| \le \|v - u\|$, with equality only when $u = P_U v$.

Proof. Write $v - u = (v - P_U v) + (P_U v - u)$. The first piece lies in $U^\perp$ and the second in U, so they are orthogonal; by the Pythagorean theorem

$$\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2 \ge \|v - P_U v\|^2,$$

with equality iff $\|P_U v - u\| = 0$, i.e. $u = P_U v$. ∎

Example 8.25 (A projection as a matrix). Onto the line $\operatorname{span}(e)$ with $e = \frac{1}{\sqrt{2}}(1, 1)$, the projection operator is the outer product $|e\rangle\langle e| = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. It satisfies $P^2 = P$ and $P^\dagger = P$—the algebraic signature of an orthogonal projection that Lecture 9 will make precise. ◁

Corollary 8.26 (Bessel's inequality). For any orthonormal list $e_1, \dots, e_m$ and any v, $\sum_{k=1}^{m} |\langle e_k, v\rangle|^2 \le \|v\|^2$.

Proof. With $U = \operatorname{span}(e_1, \dots, e_m)$, Parseval on U gives $\|P_U v\|^2 = \sum_k |\langle e_k, v\rangle|^2$, and $\|P_U v\| \le \|v\|$ since $v - P_U v \perp P_U v$ (Pythagoras again). ∎

Example 8.27 (Bessel in action). For $v = (1, 2, 2) \in \mathbb{R}^3$ and the single unit vector $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$, $|\langle e_1, v\rangle|^2 = \frac{1}{2}(1 + 2)^2 = \frac{9}{2}$, while $\|v\|^2 = 9$. Bessel's inequality $\frac{9}{2} \le 9$ holds, and the gap $9 - \frac{9}{2} = \frac{9}{2}$ is exactly the squared distance from v to the line—the part of v the single vector cannot capture. ◁

Example 8.28 (Projection onto a line and least squares). Onto the line through a unit vector e, the projection is $Pv = \langle e, v\rangle\, e$; for a general nonzero a, $Pv = \dfrac{\langle a, v\rangle}{\|a\|^2}\, a$. Least-squares fitting is this idea in disguise: to satisfy an overdetermined system $Ax = b$ as nearly as possible, project b onto range A. Best approximation says the residual is orthogonal to the range, $\langle Ay, b - Ax\rangle = 0$ for every y; rewriting the left side as $\langle y, A^\dagger(b - Ax)\rangle$—the defining property of the adjoint, stated in the Riesz remark at the end of this lecture—gives the *normal equations*

$$A^\dagger Ax = A^\dagger b,$$

whose solution minimizes $\|Ax - b\|$. ◁

Example 8.29 (Least-squares line fit). To fit $y = c_0 + c_1 x$ to the points $(0, 1), (1, 3), (2, 4)$, set up the overdetermined system $Ac = y$ with

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad y = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}.$$

The normal equations $A^\dagger Ac = A^\dagger y$ become $\left(\begin{smallmatrix} 3 & 3 \\ 3 & 5 \end{smallmatrix}\right) c = \left(\begin{smallmatrix} 8 \\ 11 \end{smallmatrix}\right)$, with solution $c = (c_0, c_1) = \left(\frac{7}{6}, \frac{3}{2}\right)$. The line $y = \frac{7}{6} + \frac{3}{2}x$ minimizes the total squared vertical error—precisely the orthogonal projection of y onto range A. ◁

Example 8.30 (A projective measurement). A spin in state $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$, measured in the $\{|0\rangle, |1\rangle\}$ basis, yields $|0\rangle$ with probability $|\langle 0, \psi\rangle|^2 = \frac{1}{2}$ and $|1\rangle$ with probability $\frac{1}{2}$. The post-measurement state is the normalized projection onto the observed eigenspace—here $|0\rangle$ or $|1\rangle$. Measurement is orthogonal projection followed by renormalization. ◁

Example 8.31 (Best polynomial fit to sin). Projecting $v(x) = \sin x$ onto the degree-≤ 5 polynomials in $C[-\pi, \pi]$ (run Gram–Schmidt on $1, x, \ldots, x^5$, then apply P_U) gives the best mean-square approximation

$$u(x) \approx 0.98786\, x - 0.15527\, x^3 + 0.00564\, x^5,$$

markedly more accurate near $\pm\pi$ than the Taylor polynomial $x - \frac{x^3}{3!} + \frac{x^5}{5!}$. Orthogonal projection, not Taylor expansion, is the right tool when uniform quality across an interval is wanted. ◁

Example 8.32 (Closest point in $\mathbb{R}^4$). To find the point of $U = \operatorname{span}\big((1, 1, 0, 0), (1, 1, 1, 2)\big)$ nearest $b = (1, 2, 3, 4)$, orthonormalize: $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0, 0)$, and since $(1, 1, 1, 2) - \langle e_1, (1, 1, 1, 2)\rangle e_1 = (0, 0, 1, 2)$, $e_2 = \frac{1}{\sqrt{5}}(0, 0, 1, 2)$. Then

$$P_U b = \langle e_1, b\rangle e_1 + \langle e_2, b\rangle e_2 = \tfrac{3}{2}(1, 1, 0, 0) + \tfrac{11}{5}(0, 0, 1, 2) = \left(\tfrac{3}{2}, \tfrac{3}{2}, \tfrac{11}{5}, \tfrac{22}{5}\right),$$

the least-squares closest point. ◁

Example 8.33 (Projection onto a plane). With the orthonormal e_1, e_2 of Example 8.16, the projection onto that plane is $P_U = |e_1\rangle\langle e_1| + |e_2\rangle\langle e_2|$. Acting on $b = (1, 0, 0)$,

$$P_U b = \langle e_1, b\rangle e_1 + \langle e_2, b\rangle e_2 = \tfrac{2}{5}(2, 0, 1) + \tfrac{1}{30}(1, 5, -2) = \tfrac{1}{6}(5, 1, 2),$$

and the residual $b - P_U b = \frac{1}{6}(1, -1, -2)$ is parallel to the plane's normal $(1, -1, -2)$, as the orthogonal decomposition demands. ◁

4 The Riesz representation theorem

A *linear functional* on V is a linear map $\varphi\colon V \to \mathbb{C}$. In an inner product space every functional is "take the inner product with a fixed vector."

> **Theorem 8.34 (Riesz representation).** Let V be finite-dimensional. For every linear functional φ on V there is a unique $w \in V$ with
>
> $$\varphi(v) = \langle w, v\rangle \qquad \text{for all } v \in V.$$

Proof. Fix an orthonormal basis $e_1, \dots, e_n$ and set $w = \sum_k \overline{\varphi(e_k)}\, e_k$. Then for any $v = \sum_k \langle e_k, v\rangle e_k$,

$$\langle w, v\rangle = \sum_k \varphi(e_k)\langle e_k, v\rangle = \varphi\Big(\sum_k \langle e_k, v\rangle e_k\Big) = \varphi(v),$$

using $\langle w, e_k\rangle = \varphi(e_k)$ in the first step and linearity of φ in the second. For uniqueness, if $\langle w, v\rangle = \langle w', v\rangle$ for all v then $\langle w - w', v\rangle = 0$ for all v; taking $v = w - w'$ forces $w = w'$. ∎

Remark 8.35 (Bra–ket duality). Riesz says exactly that the map $|w\rangle \mapsto \langle w|$ is a bijection: every *bra* (linear functional φ) is $\langle w|$ for a unique *ket* $|w\rangle$. The bra space is thus an exact (conjugate-linear) copy of the ket space—the rigorous content of Dirac's notation. The conjugation $w = \sum_k \overline{\varphi(e_k)} e_k$ is the same conjugation that makes $|\phi\rangle \mapsto \langle\phi|$ antilinear.

Example 8.36 (A represented functional). On $\mathbb{C}^2$ the functional $\varphi(v) = v_1 + 2iv_2$ is $\langle w, v\rangle$ with $w = (\overline{1}, \overline{2i}) = (1, -2i)$, since $\langle w, v\rangle = \overline{1}\, v_1 + \overline{(-2i)} v_2 = v_1 + 2iv_2$. The conjugation lands the coefficients in the right slot. ◁

Example 8.37 (Riesz on a function space). "Evaluate at 0" is a linear functional on $\mathcal{P}_n$ with the L^2 inner product, so by Riesz it equals $\langle w, \cdot\rangle$ for a fixed polynomial w—a *reproducing kernel*: $p(0) = \langle w, p\rangle$ for all p. Every linear functional on a finite-dimensional inner product space conceals such a representing vector. ◁

Example 8.38 (The trace as an inner product). On $M_n(\mathbb{C})$ with the Frobenius product, the trace functional $\varphi(A) = \operatorname{tr} A$ is represented by the identity matrix: $\langle I, A\rangle = \operatorname{tr}(I^\dagger A) = \operatorname{tr} A$, so $w = I$. Even "sum the diagonal" is secretly an inner product with a fixed vector. ◁

Remark 8.39 (From Riesz to the adjoint). Riesz is the engine of the next lecture. Given $T \in \mathcal{L}(V)$ and a fixed u, the map $v \mapsto \langle u, Tv\rangle$ is a linear functional, so by Riesz it equals $\langle w, v\rangle$ for a unique w; defining $T^\dagger u = w$ produces the *adjoint*, characterized by $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$. The mere existence of adjoints is Riesz in disguise.

5 Measurement, expansion, and what comes next

Orthonormal bases are the language of measurement. The possible outcomes of an observable correspond to an orthonormal eigenbasis $\{|e_k\rangle\}$; a state $|\psi\rangle = \sum_k \langle e_k, \psi\rangle |e_k\rangle$ yields outcome k with probability $|\langle e_k, \psi\rangle|^2$, and Parseval guarantees these sum to $\|\psi\|^2 = 1$. Measurement *projects* the state onto the observed eigenspace—exactly the orthogonal projection P_U—and the completeness relation $\sum_k |e_k\rangle\langle e_k| = I$ is what lets us insert a basis anywhere in a calculation.

Example 8.40 (Probabilities sum to one). If $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}\,|1\rangle$, the outcome probabilities are $|\langle 0, \psi\rangle|^2 = \frac{1}{3}$ and $|\langle 1, \psi\rangle|^2 = \frac{2}{3}$, summing to 1 precisely because Parseval gives $\|\psi\|^2 = \frac{1}{3} + \frac{2}{3} = 1$. Normalization of a state *is* Parseval's relation. ◁

Example 8.41 (Expanding in an energy eigenbasis). If $\{|E_n\rangle\}$ is an orthonormal basis of energy eigenstates, then $|\psi\rangle = \sum_n \langle E_n, \psi\rangle |E_n\rangle$ and the time-evolved state is $|\psi(t)\rangle = \sum_n e^{-iE_n t/\hbar}\langle E_n, \psi\rangle |E_n\rangle$ (Lecture 6): each coordinate merely rotates by a phase. The probabilities $|\langle E_n, \psi\rangle|^2$ are unchanged in time, expressing conservation of energy. ◁

Remark 8.42 (The infinite-dimensional caveat). In infinite dimensions Bessel's inequality can be strict and the decomposition $V = U \oplus U^\perp$ can fail for a non-closed U: an orthonormal *list* need not be a basis. Completeness is the remedy, upgrading Bessel to Parseval and restoring these theorems—the business of Lecture 13.

The remaining ingredient is the adjoint. We have seen (Riesz) that inner products turn vectors into functionals; next lecture we turn *operators* into their adjoints $T^\dagger$ via $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$, and single out the self-adjoint, unitary, and normal operators—the ones whose eigenvectors form an orthonormal basis, by the spectral theorem of Lecture 10.

Summary of main concepts

An orthonormal basis satisfies $\langle e_j, e_k\rangle = \delta_{jk}$; in it coordinates are inner products, $v = \sum_k \langle e_k, v\rangle e_k$, the norm obeys Parseval $\|v\|^2 = \sum_k |\langle e_k, v\rangle|^2$, and $\sum_k |e_k\rangle\langle e_k| = I$. The Gram–Schmidt procedure builds such a basis from any basis, preserving partial spans, and shows every finite-dimensional inner product space has an ONB (and every orthonormal list extends to one). The orthogonal projection $P_U v = \sum_k \langle e_k, v\rangle e_k$ gives the orthogonal decomposition $V = U \oplus U^\perp$ and solves the best-approximation problem, with Bessel's inequality as a corollary. The Riesz theorem represents every linear functional as $\langle w, \cdot\rangle$, which is precisely the duality between bras and kets.

- **Orthonormal basis**—$\langle e_j, e_k\rangle = \delta_{jk}$; $v = \sum_k \langle e_k, v\rangle e_k$, Parseval, $\sum_k |e_k\rangle\langle e_k| = I$.
- **Gram–Schmidt**—builds an orthonormal basis from any basis while preserving partial spans.
- **Orthogonal projection and decomposition**—$V = U \oplus U^\perp$; $P_U v$ is the best approximation; Bessel's inequality.
- **Riesz representation**—every functional is $\langle w, \cdot\rangle$—the duality between bras and kets.

Exercises for this lecture are in the companion sheet `exercises-8.pdf`.